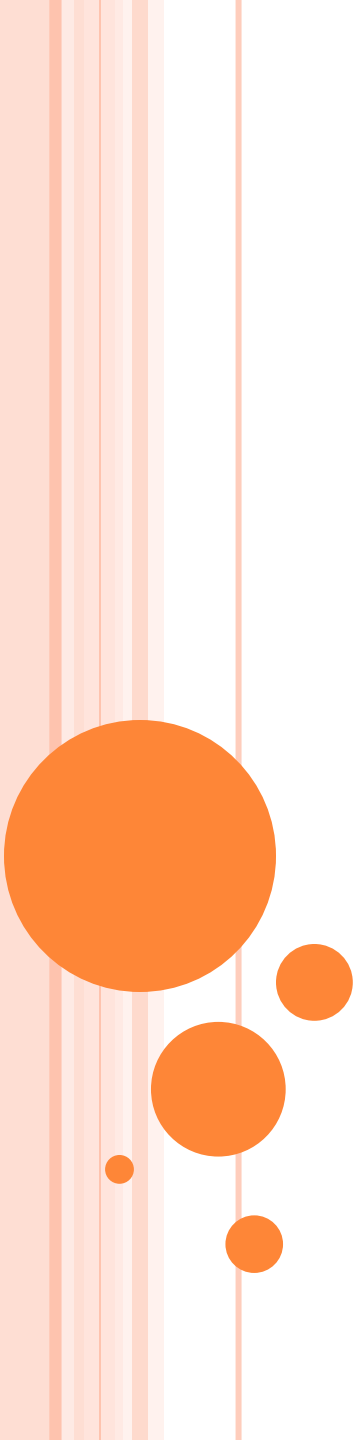




GRAPH THEORY

SHORTEST PATHS II

BELLMAN-FORD

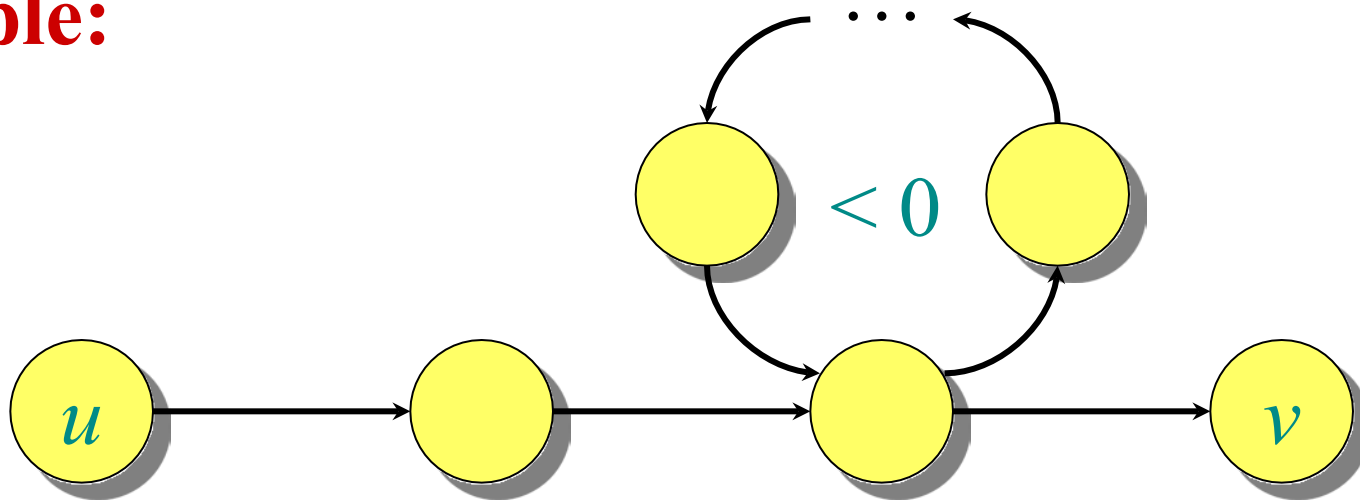
ALGORITHM

Design and Analysis of Algorithms

NEGATIVE-WEIGHT CYCLES

Recall: If a graph $G = (V, E)$ contains a negative-weight cycle, then some shortest paths may not exist.

Example:

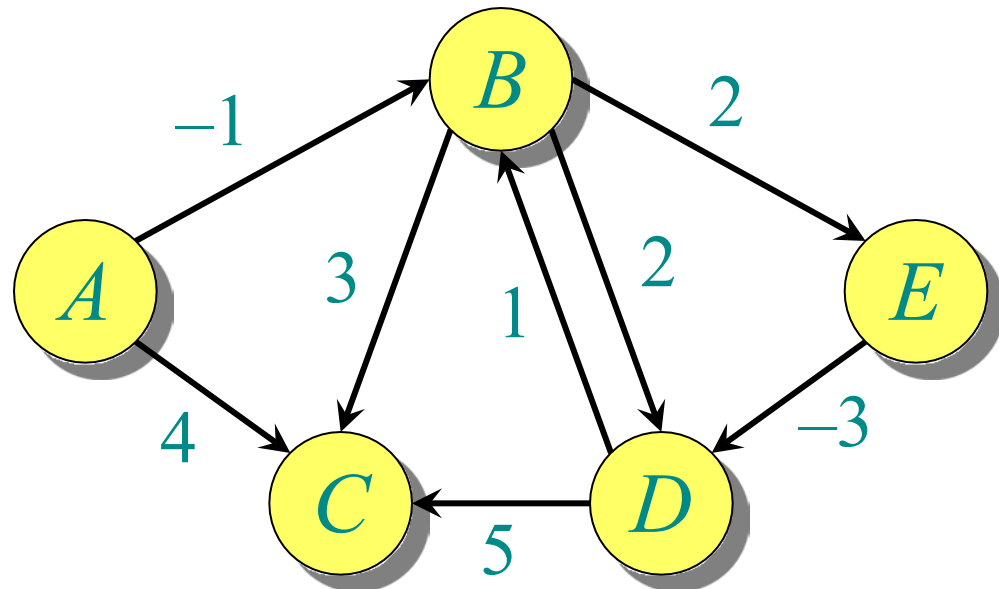


Bellman-Ford algorithm: Finds all shortest-path lengths from a ***source*** $s \in V$ to all $v \in V$ or determines that a negative-weight cycle exists.

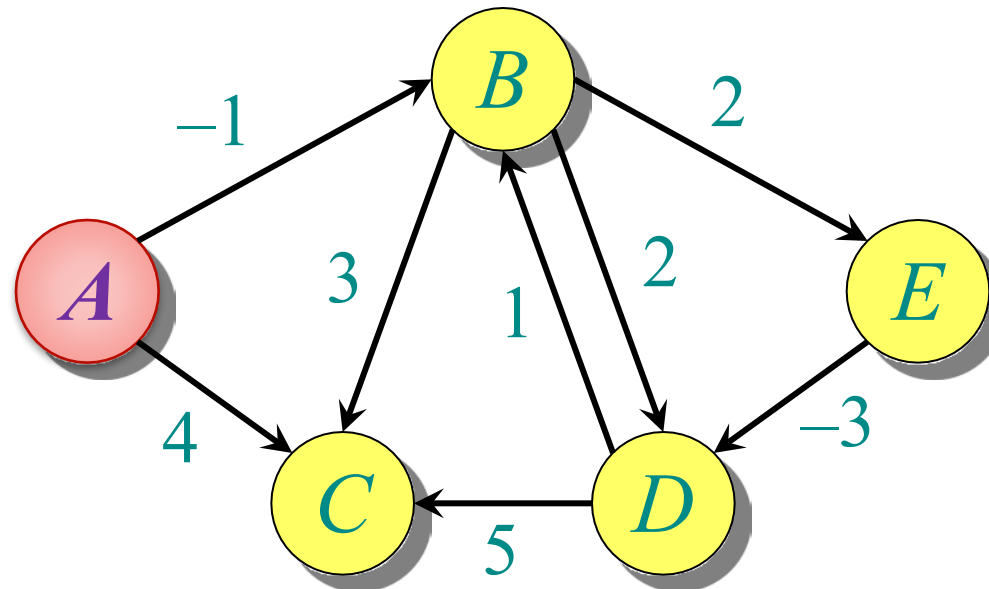
BELLMAN-FORD ALGORITHM

```
 $d[s] \leftarrow 0$   
for each  $v \in V - \{s\}$   
    do  $d[v] \leftarrow \infty$  } initialization  
  
for  $i \leftarrow 1$  to  $|V| - 1$   
    do for each edge  $(u, v) \in E$   
        do if  $d[v] > d[u] + w(u, v)$   
            then  $d[v] \leftarrow d[u] + w(u, v)$  } relaxation step  
  
for each edge  $(u, v) \in E$   
    do if  $d[v] > d[u] + w(u, v)$   
        then report that a negative-weight cycle exists  
  
At the end,  $d[v] = \delta(s, v)$ , if no negative-weight cycles.  
Time =  $O(VE)$ .
```

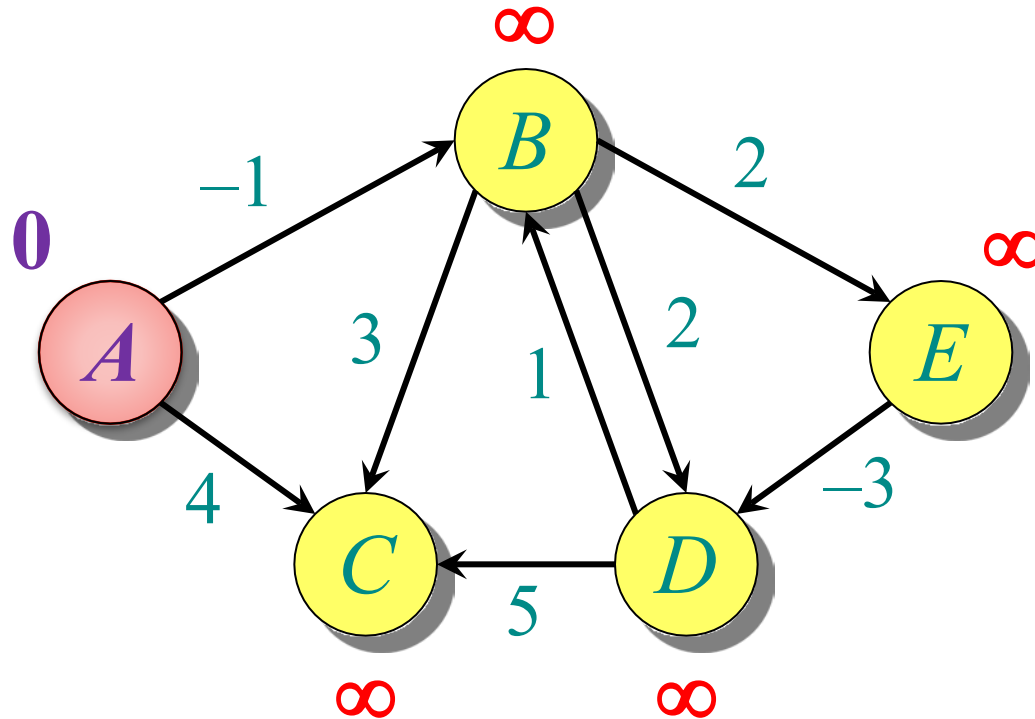
EXAMPLE OF BELLMAN-FORD



EXAMPLE OF BELLMAN-FORD

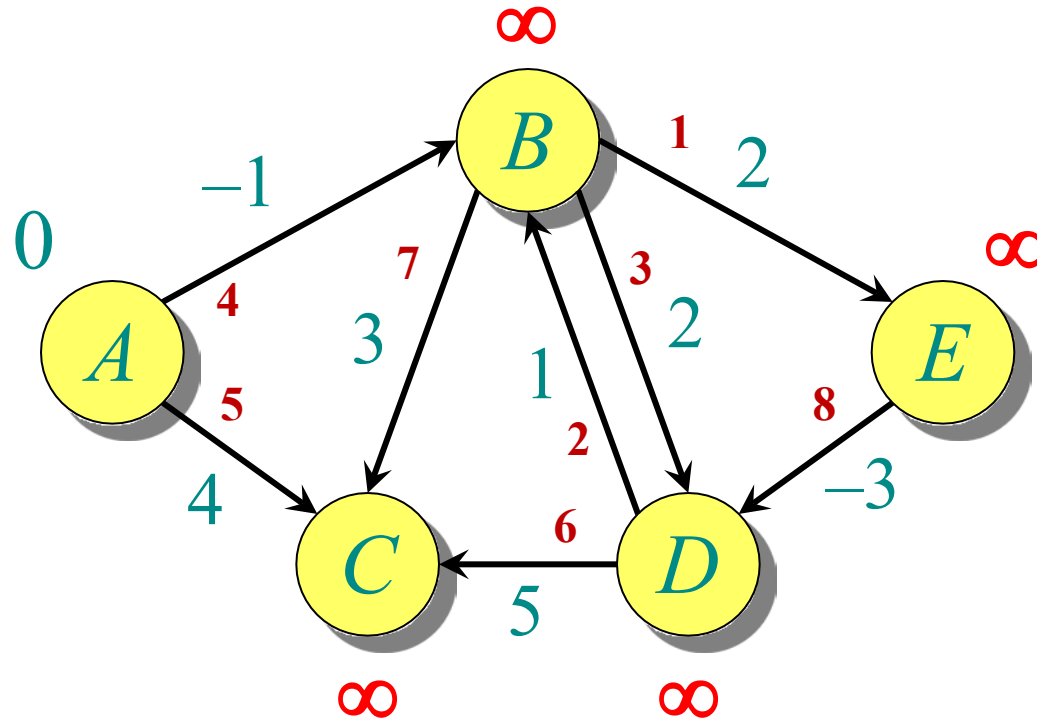


EXAMPLE OF BELLMAN-FORD



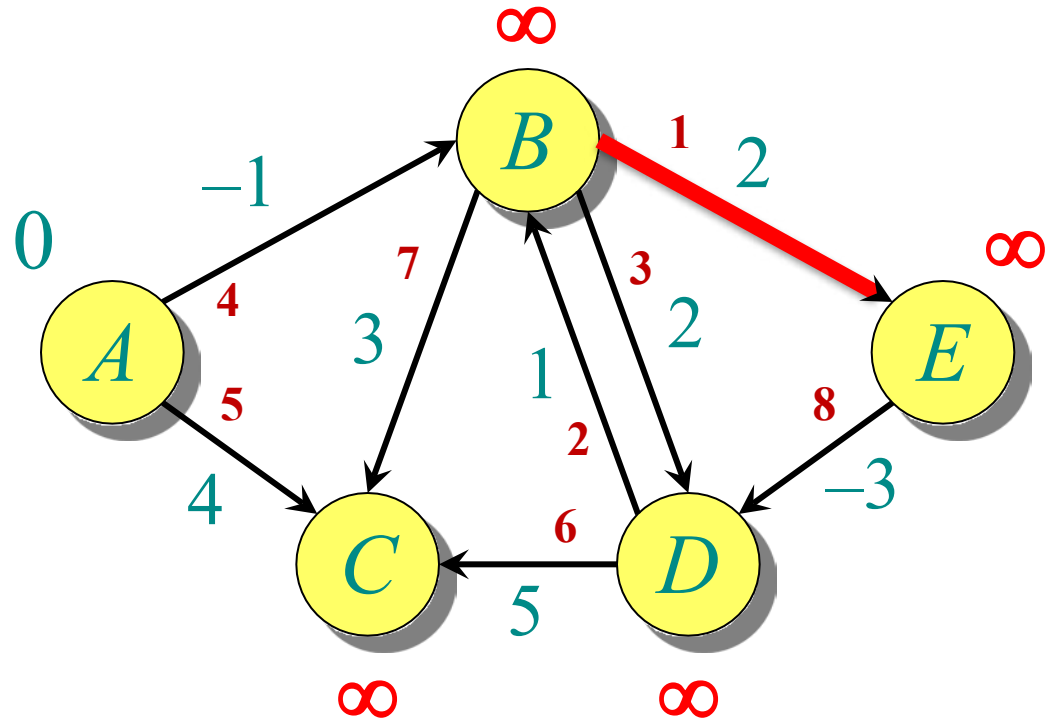
Initialization.

EXAMPLE OF BELLMAN-FORD

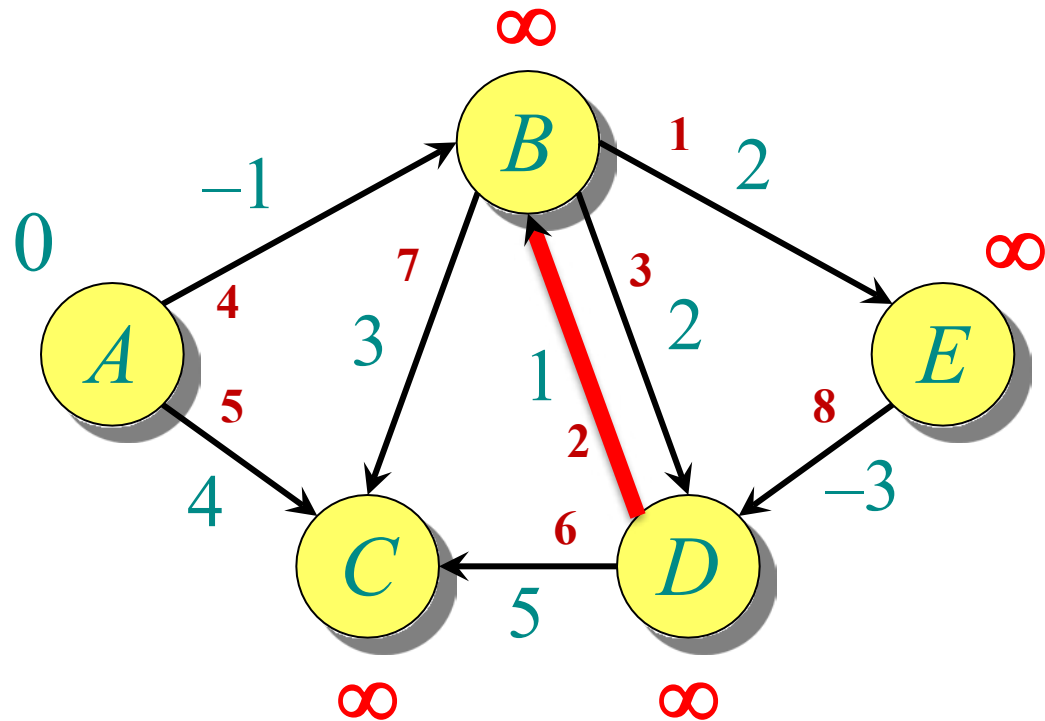


Note: Red number with edges is showing order of edge relation

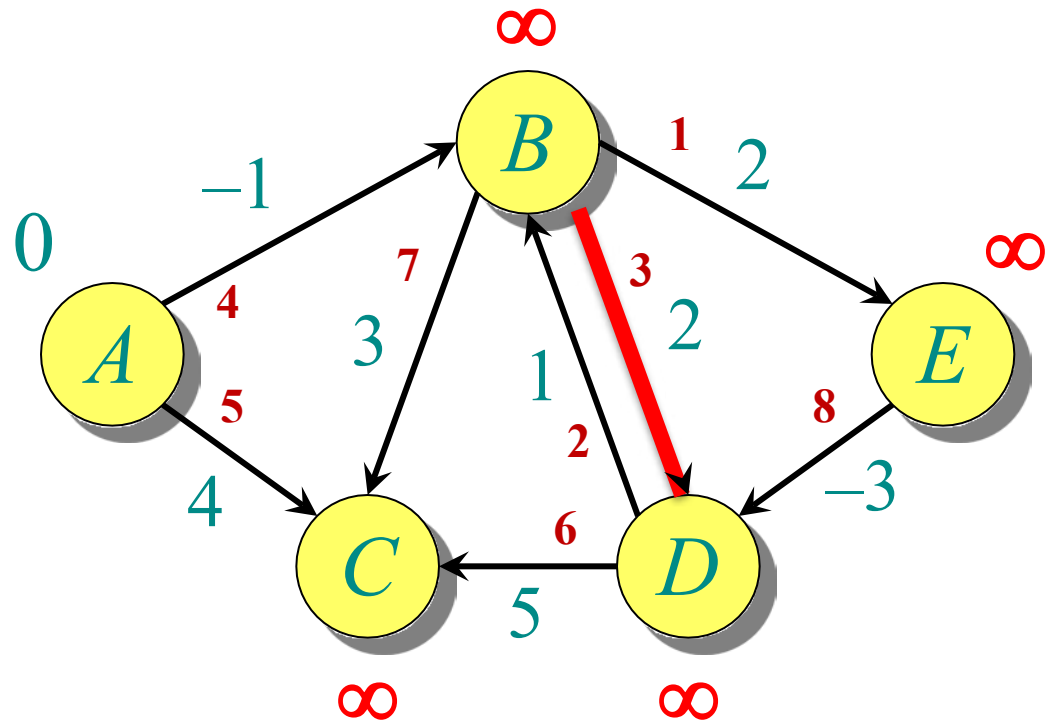
EXAMPLE OF BELLMAN-FORD



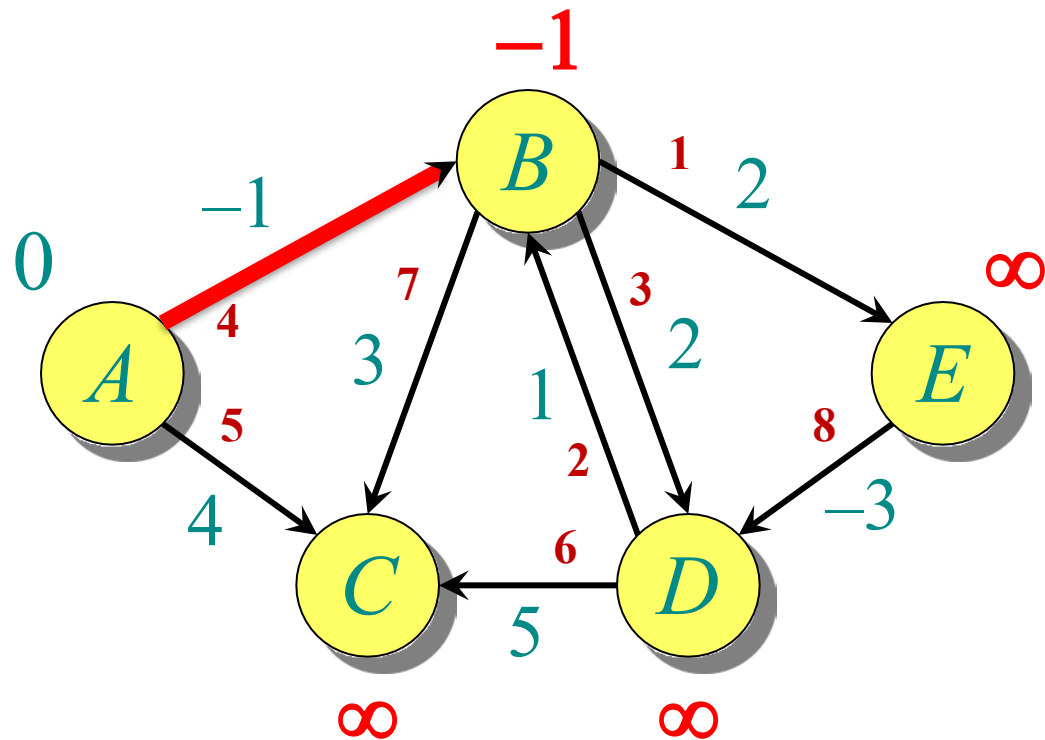
EXAMPLE OF BELLMAN-FORD



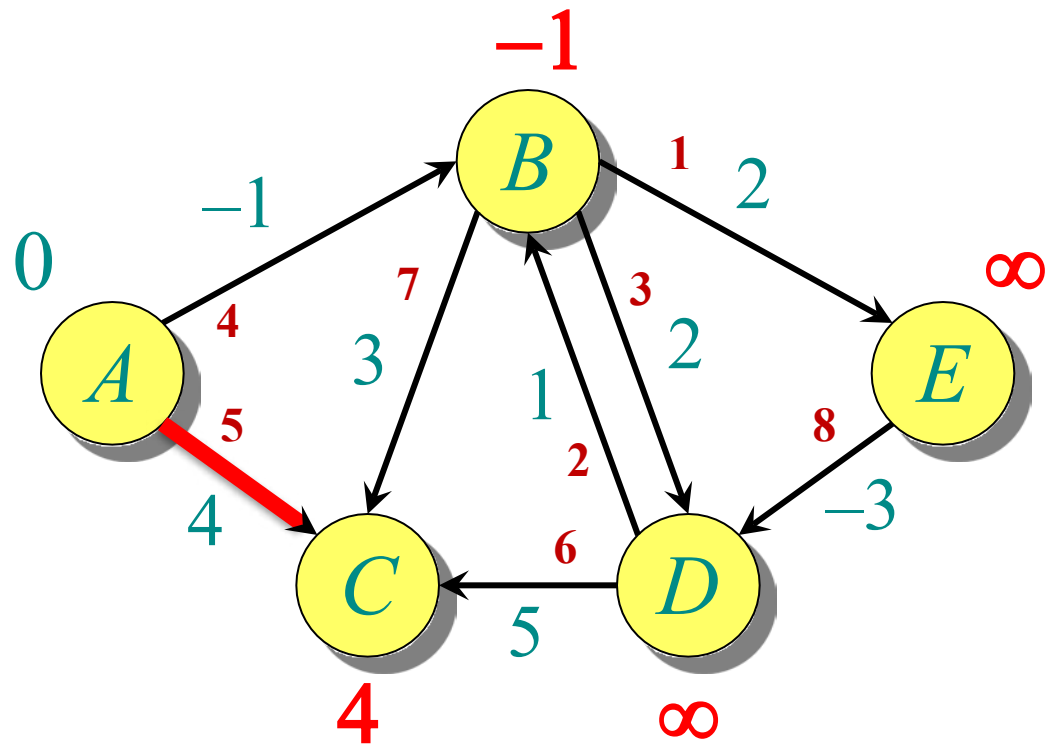
EXAMPLE OF BELLMAN-FORD



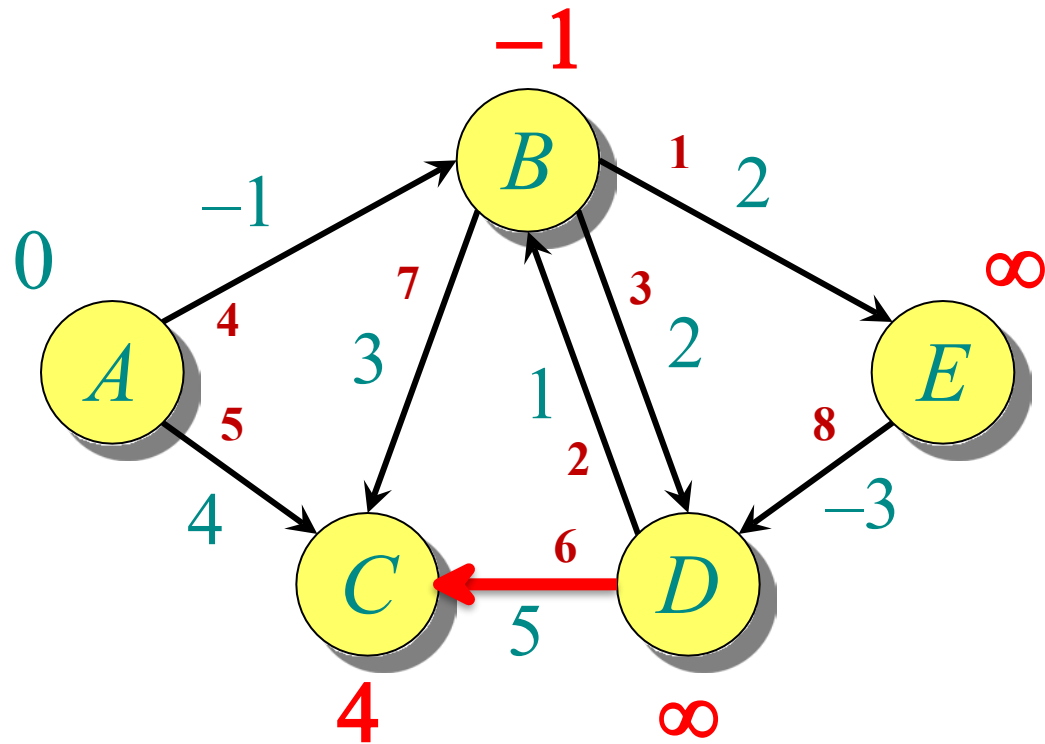
EXAMPLE OF BELLMAN-FORD



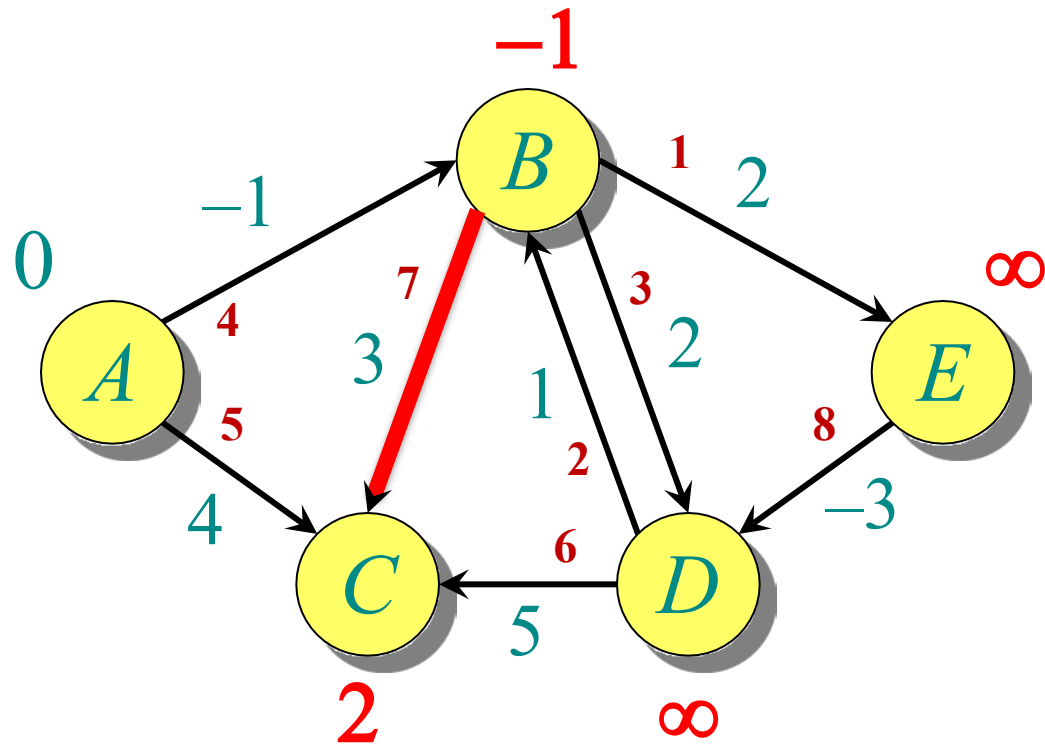
EXAMPLE OF BELLMAN-FORD



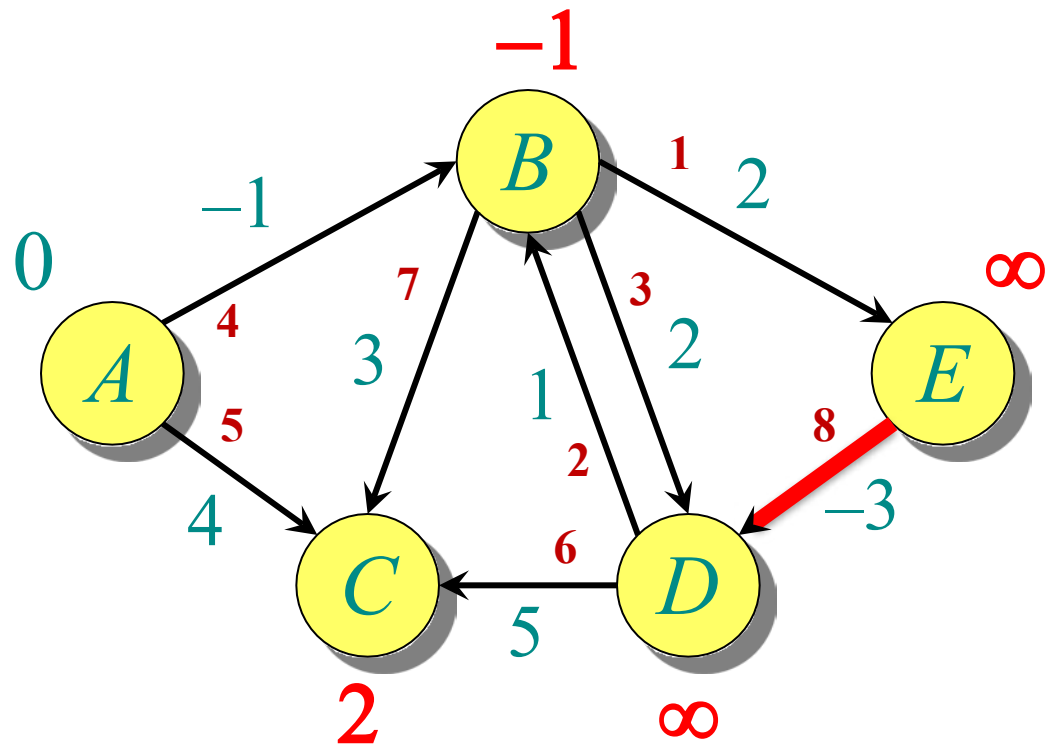
EXAMPLE OF BELLMAN-FORD



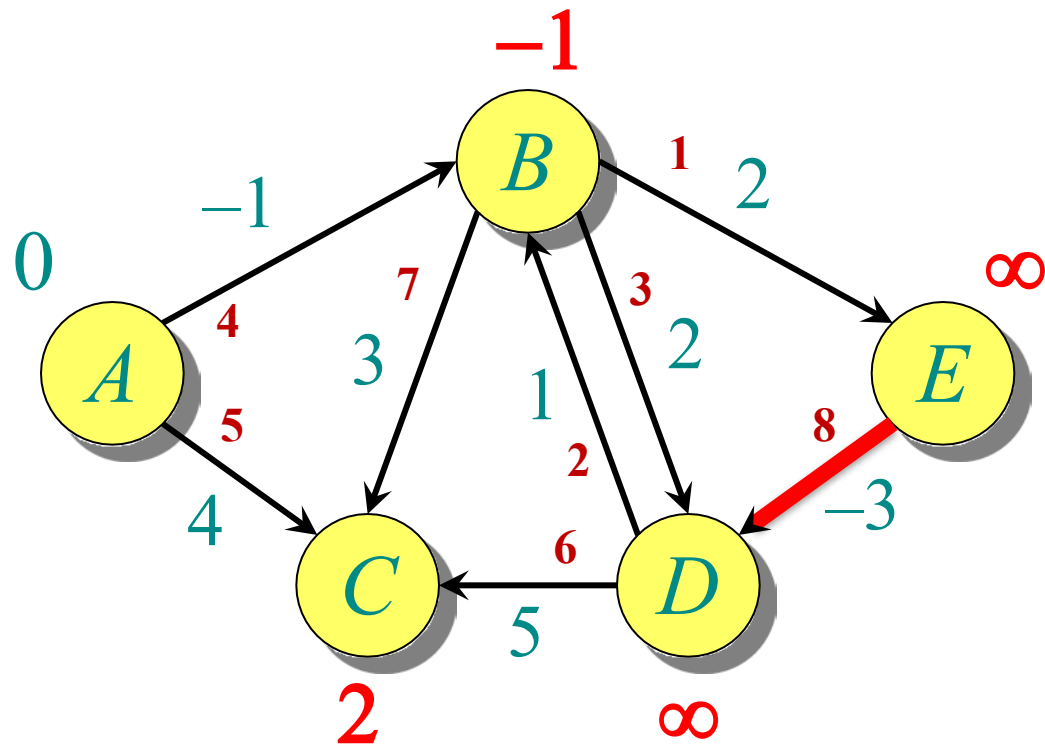
EXAMPLE OF BELLMAN-FORD



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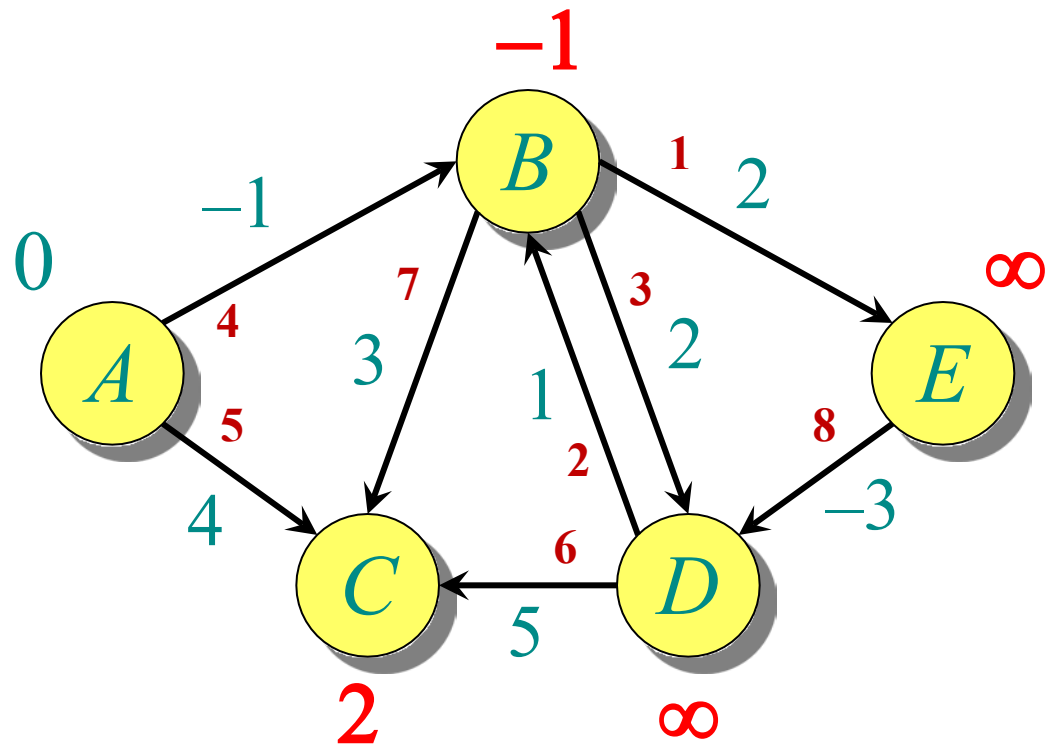


EXAMPLE OF BELLMAN-FORD



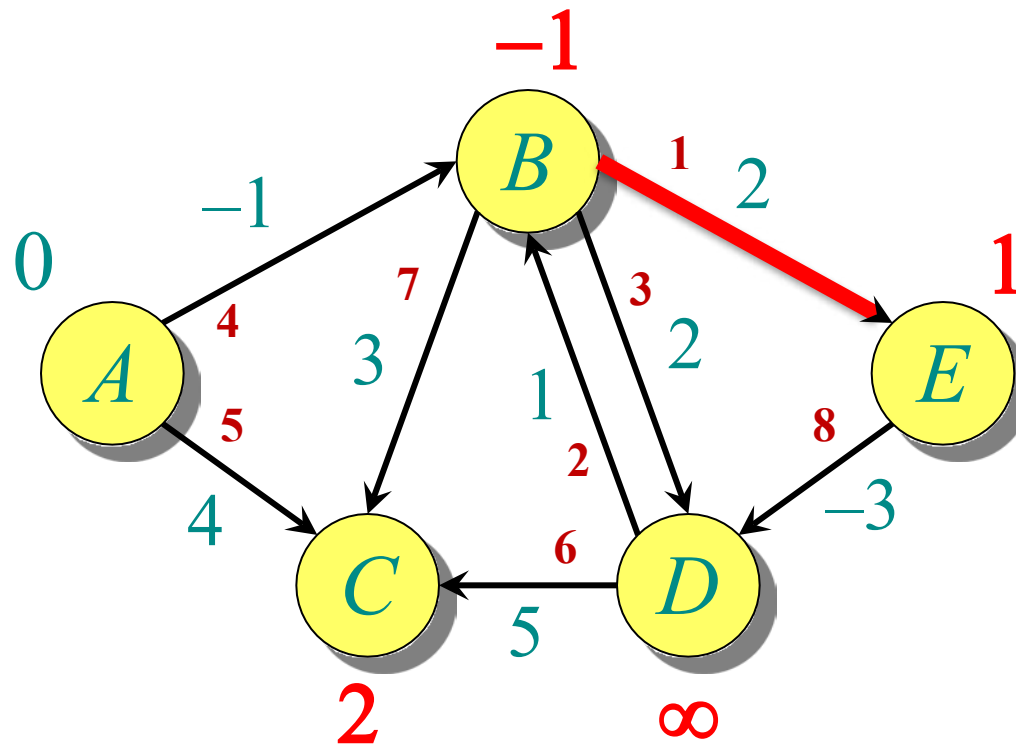
End of Pass 1

EXAMPLE OF BELLMAN-FORD



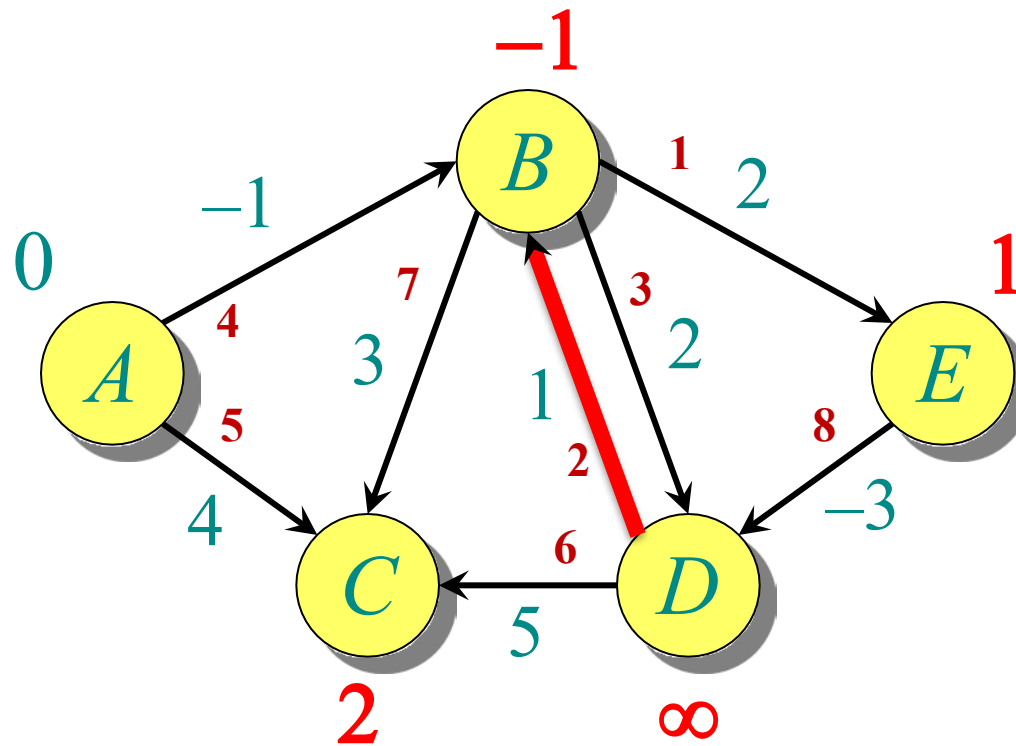
End of Pass 1

EXAMPLE OF BELLMAN-FORD-PASS2

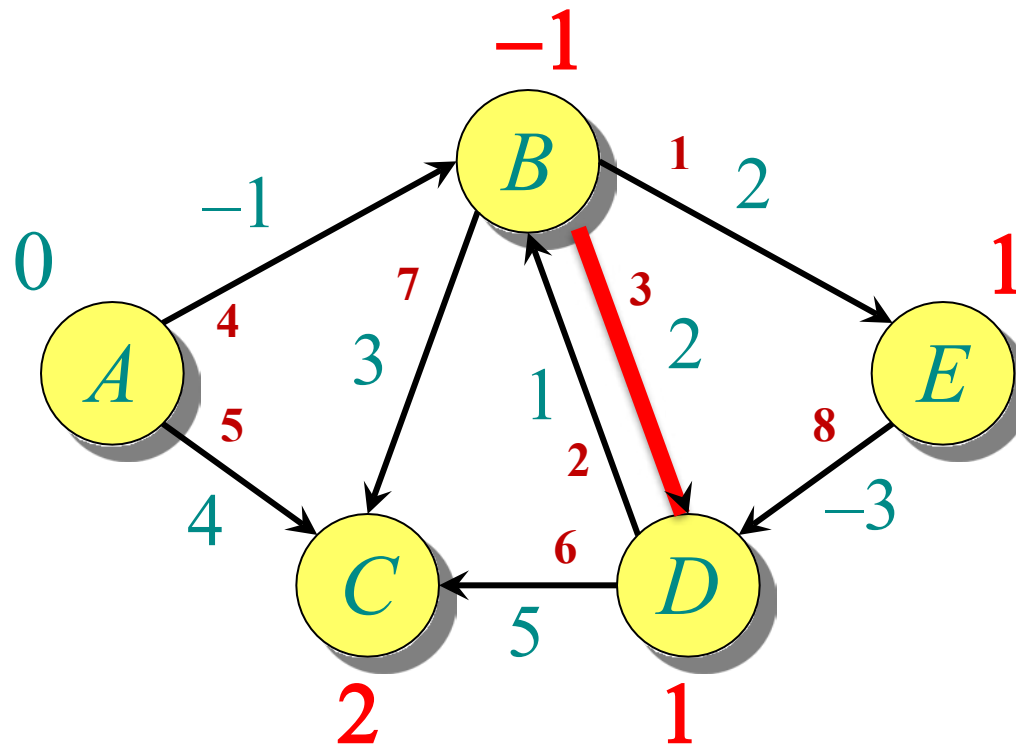


Note: Red number with edges is showing order of edge relation

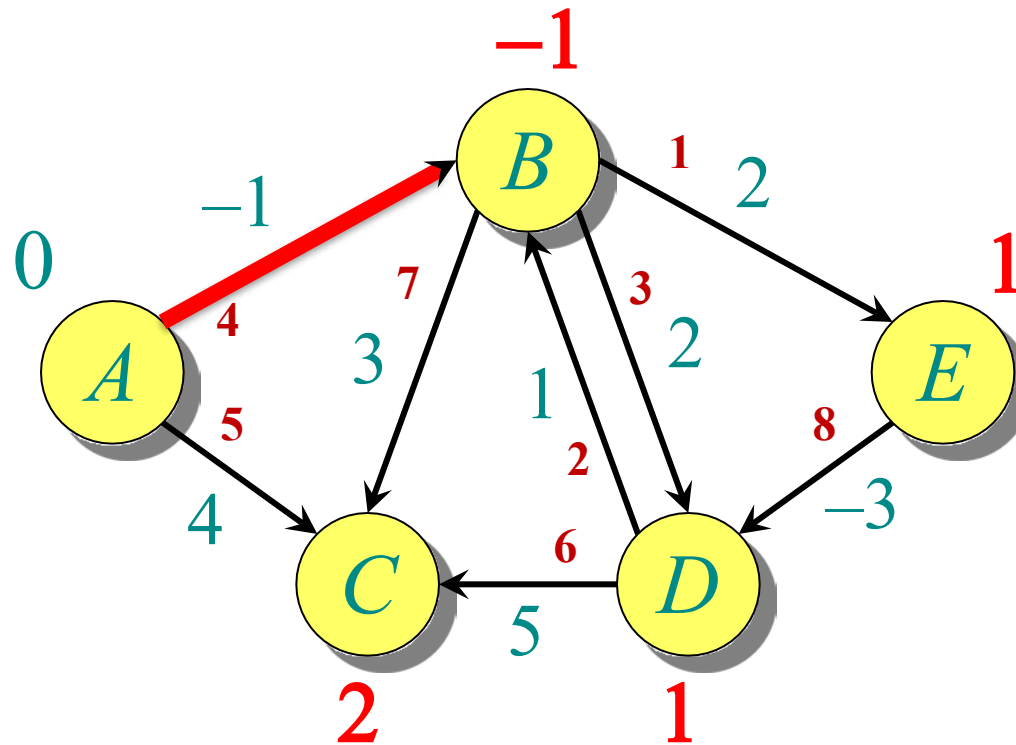
EXAMPLE OF BELLMAN-FORD



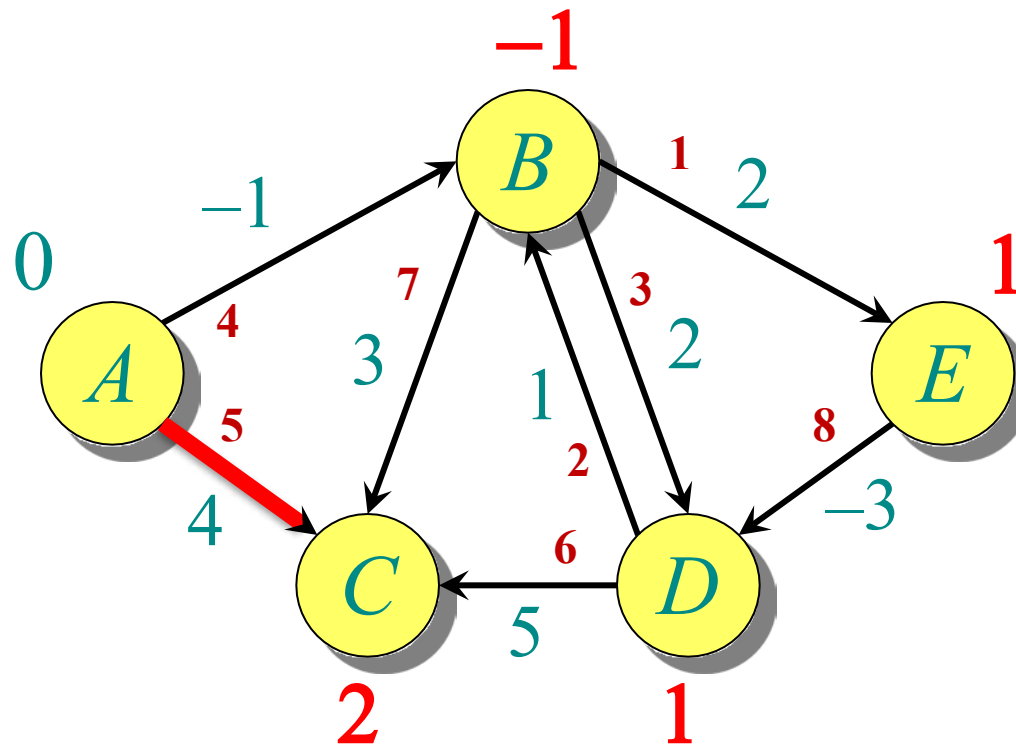
EXAMPLE OF BELLMAN-FORD



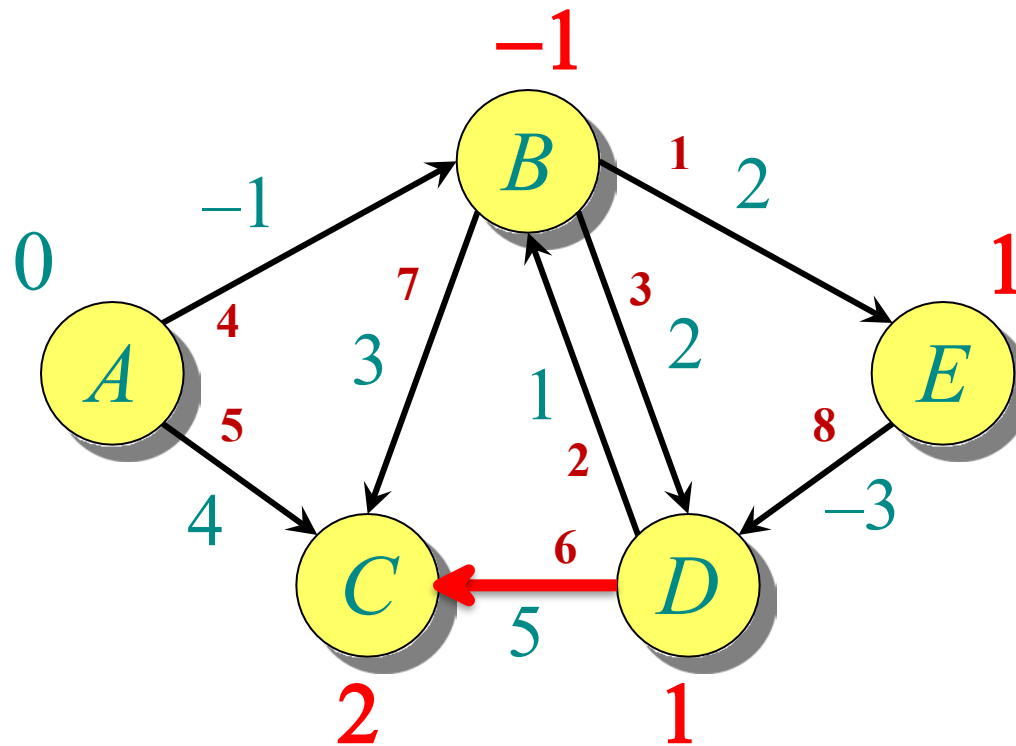
EXAMPLE OF BELLMAN-FORD



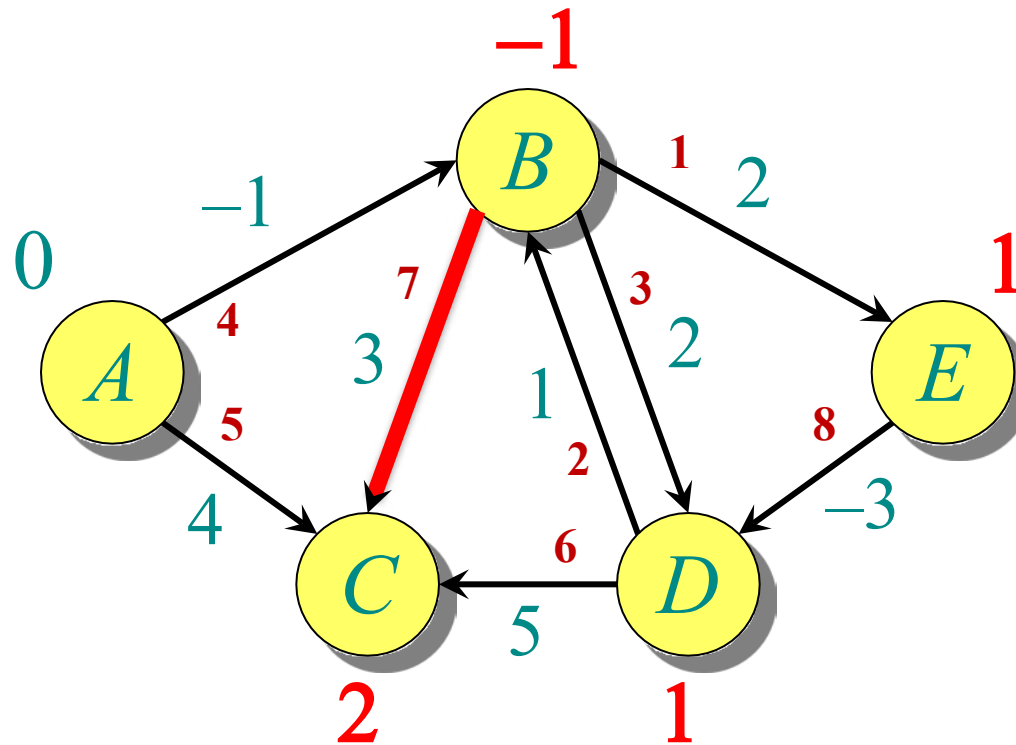
EXAMPLE OF BELLMAN-FORD



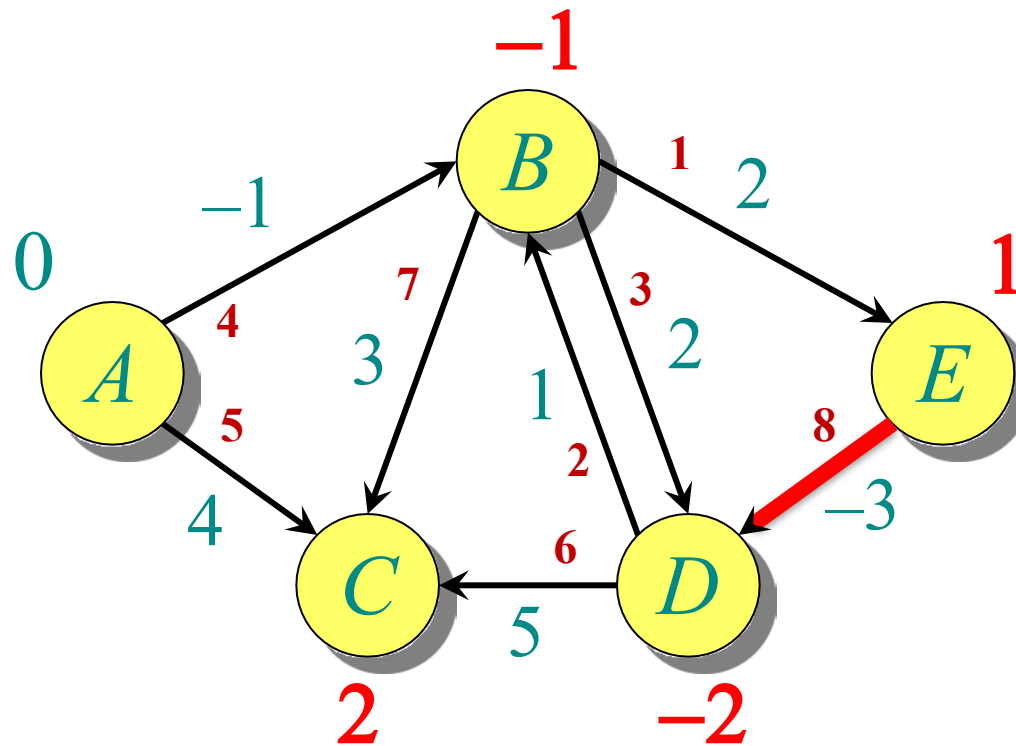
EXAMPLE OF BELLMAN-FORD



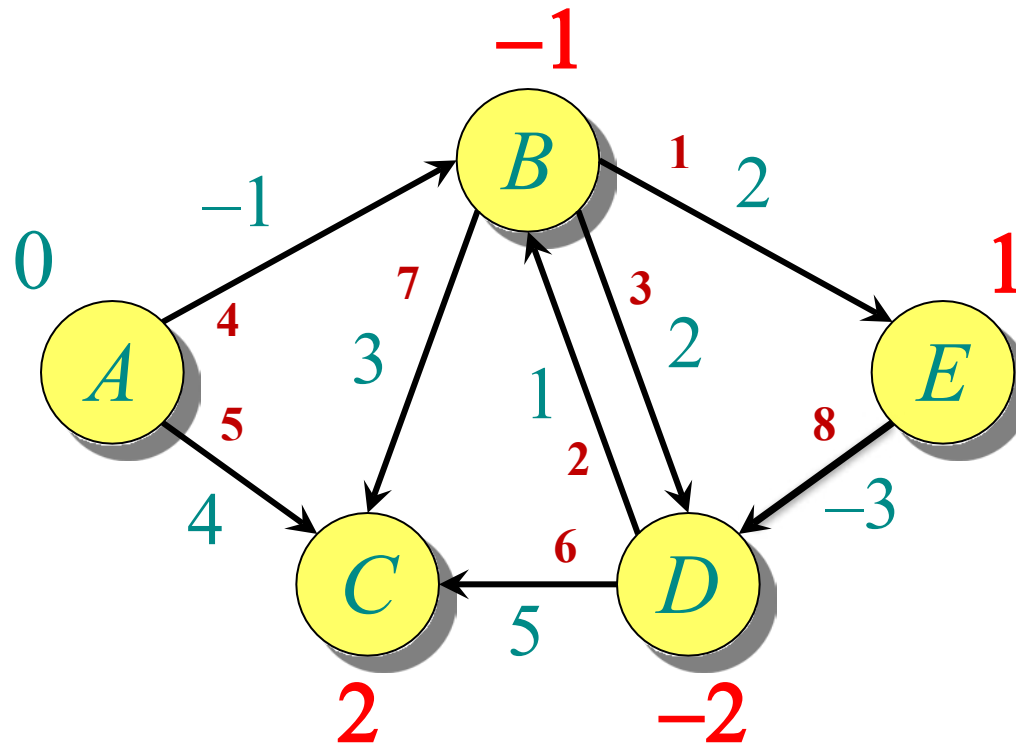
EXAMPLE OF BELLMAN-FORD



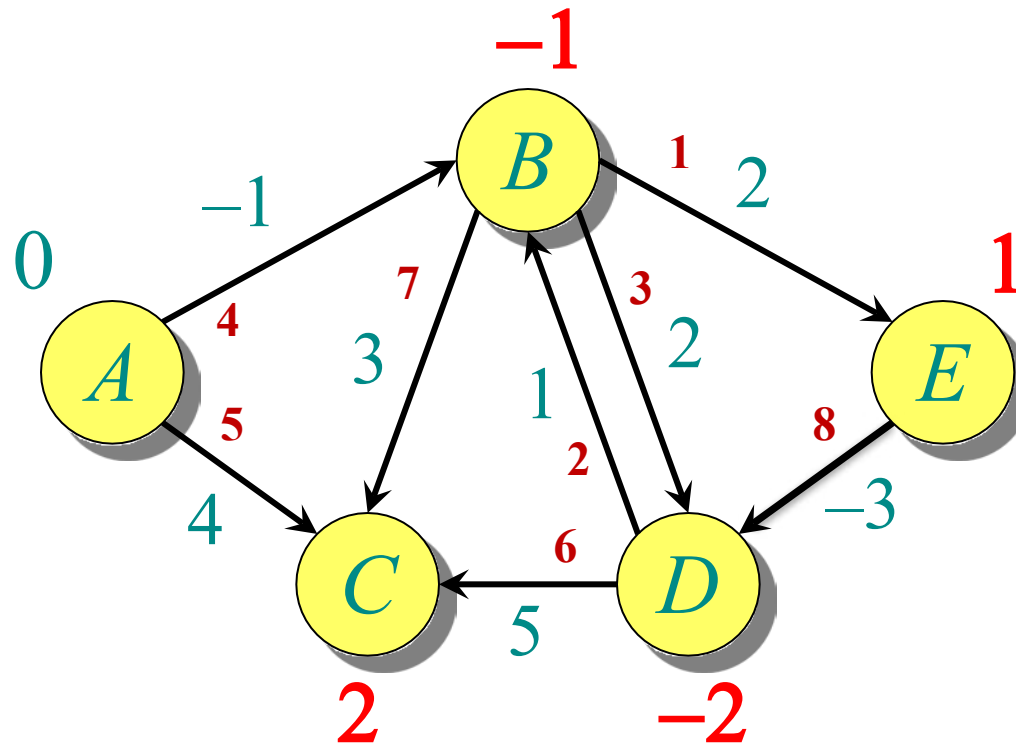
EXAMPLE OF BELLMAN-FORD



EXAMPLE OF BELLMAN-FORD



EXAMPLE OF BELLMAN-FORD



End of pass 2 (and 3 and 4).

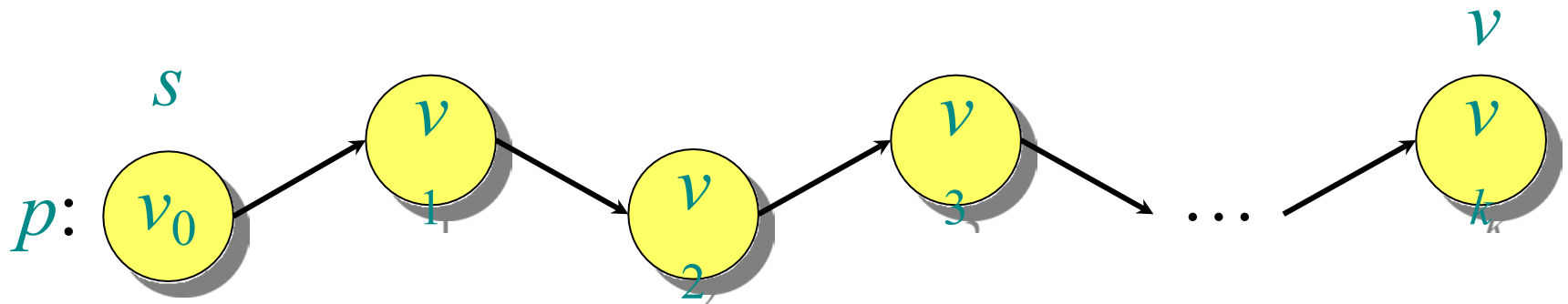
Correctness

Theorem. If $G = (V, E)$ contains no negative-weight cycles, then after the Bellman-Ford algorithm executes, $d[v] = \delta(s, v)$ for all $v \in V$.

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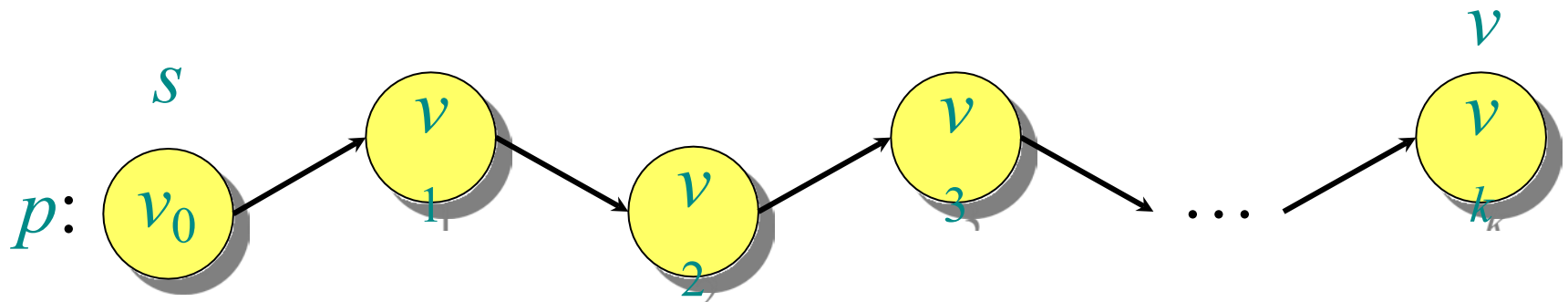
Proof. Let $v \in V$ be any vertex, and consider a shortest path p from s to v with the minimum number of edges.



Since p is a shortest path, we have

$$\delta(s, v_i) = \delta(s, v_{i-1}) + w(v_{i-1}, v_i) .$$

CORRECTNESS (CONTINUED)



Initially, $d[v_0] = 0 = \delta(s, v_0)$, and $d[v_0]$ is unchanged by subsequent relaxations (because of the lemma from *Shortest Paths I* that $d[v] \geq \delta(s, v)$).

- After 1 pass through E , we have $d[v_1] = \delta(s, v_1)$.
- After 2 passes through E , we have $d[v_2] = \delta(s, v_2)$.
- After k passes through E , we have $d[v_k] = \delta(s, v_k)$.

Since G contains no negative-weight cycles, p is simple. Longest simple path has $\leq |V| - 1$ edges. $\square$

DETECTION OF NEGATIVE-WEIGHT CYCLES

Corollary. If a value $d[v]$ fails to converge after $|V| - 1$ passes, there exists a negative-weight cycle in G reachable from s . 

SHORTEST PATHS

Single-source shortest paths

- Nonnegative edge weights
 - Dijkstra's algorithm: $O(E + V \lg V)$
- General
 - Bellman-Ford algorithm: $O(VE)$
- DAG
 - One pass of Bellman-Ford: $O(V + E)$

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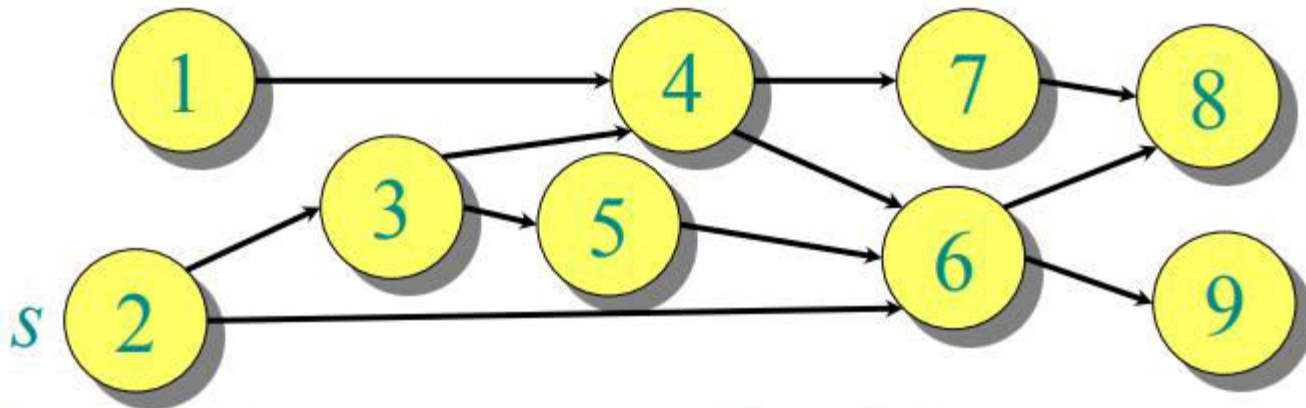
All-pairs shortest paths

- Nonnegative edge weights
 - Dijkstra's algorithm $|V|$ times: $O(VE + V^2 \lg V)$

DIRECTED ACYCLIC GRAPH (DAG)

If the graph is a *directed acyclic graph* (**DAG**), we first *topologically sort* the vertices.

- Determine $f: V \rightarrow \{1, 2, \dots, |V|\}$ such that $(u, v) \in E \Rightarrow f(u) < f(v)$.
- $O(V + E)$ time using depth-first search.



Walk through the vertices $u \in V$ in this order, relaxing the edges in $Adj[u]$, thereby obtaining the shortest paths from s in a total of $O(V + E)$ time.

ALL-PAIRS SHORTEST PATHS

Input: Digraph $G = (V, E)$, where $V = \{1, 2, \dots, n\}$, with edge-weight function $w : E \rightarrow \mathbb{R}$.

Output: $n \times n$ matrix of shortest-path lengths $\delta(i, j)$ for all $i, j \in V$.

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IDEA:

- Run Bellman-Ford once from each vertex.
- Time = $O(V^2E)$.
- Dense graph ($\Theta(n^2)$ edges) $\Rightarrow \Theta(n^4)$ time in the worst case.

Good first try!

REFERENCE

○ Introduction to Algorithms

- Single Source Shortest Path
- Chapter # 24
- Thomas H. Cormen